



grindTryThisgrind

satunknownunknown

leant?leantz3?grindLibrarySuggestions

```
grind
TryThis
LibrarySuggestions
bv_decide
```

grindTryThis

grind
grind

solve?

leant?

solve?

PropBool

leant?

grindomegagrobnerbv_decide

leant?leant

refine

solve?

unsat

Leant.Tactic

```
import Leant.Tactic

example (A B C : Prop)
  (f : A -> B -> C) (g : A -> B) (a : A) : C := by
  leant?
  -- Try this:
```

```
-- exact f a (g a)
```

TryThis

0160237e4bdf

1d2dde6d64cb

0bd679a404be

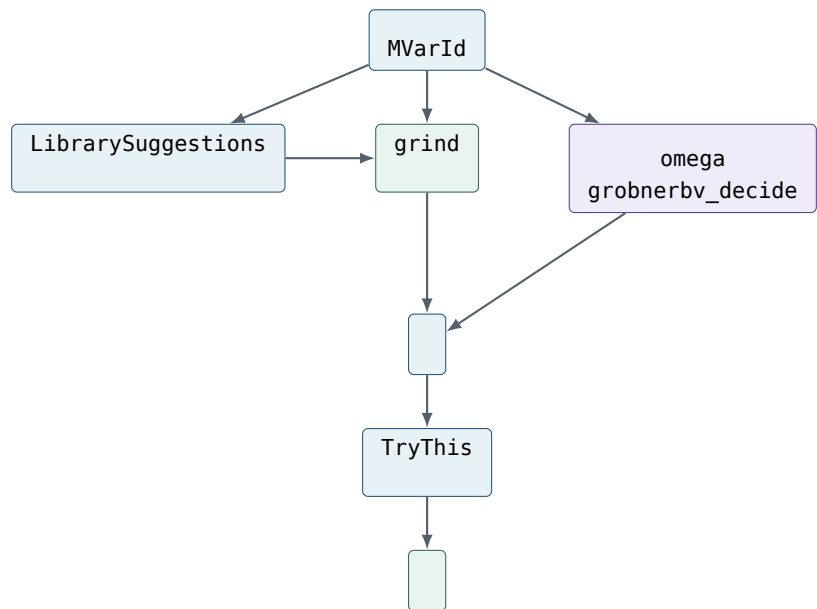
dab7a1518f51

f50b5eac8672

a4ed29b0b9fd

grindgrind?bv_decidegrindbv_decide

0160237e4bdf



grind

grind

grind?

LibrarySuggestions.select"grind"

bv_decidegrindgrind

grind

grind

TryThis

Lean.Meta.Tactic.TryThisexactrefineexpose_names

leant?refine

z3?

solve?

exactrefineTryThis

LibrarySuggestions

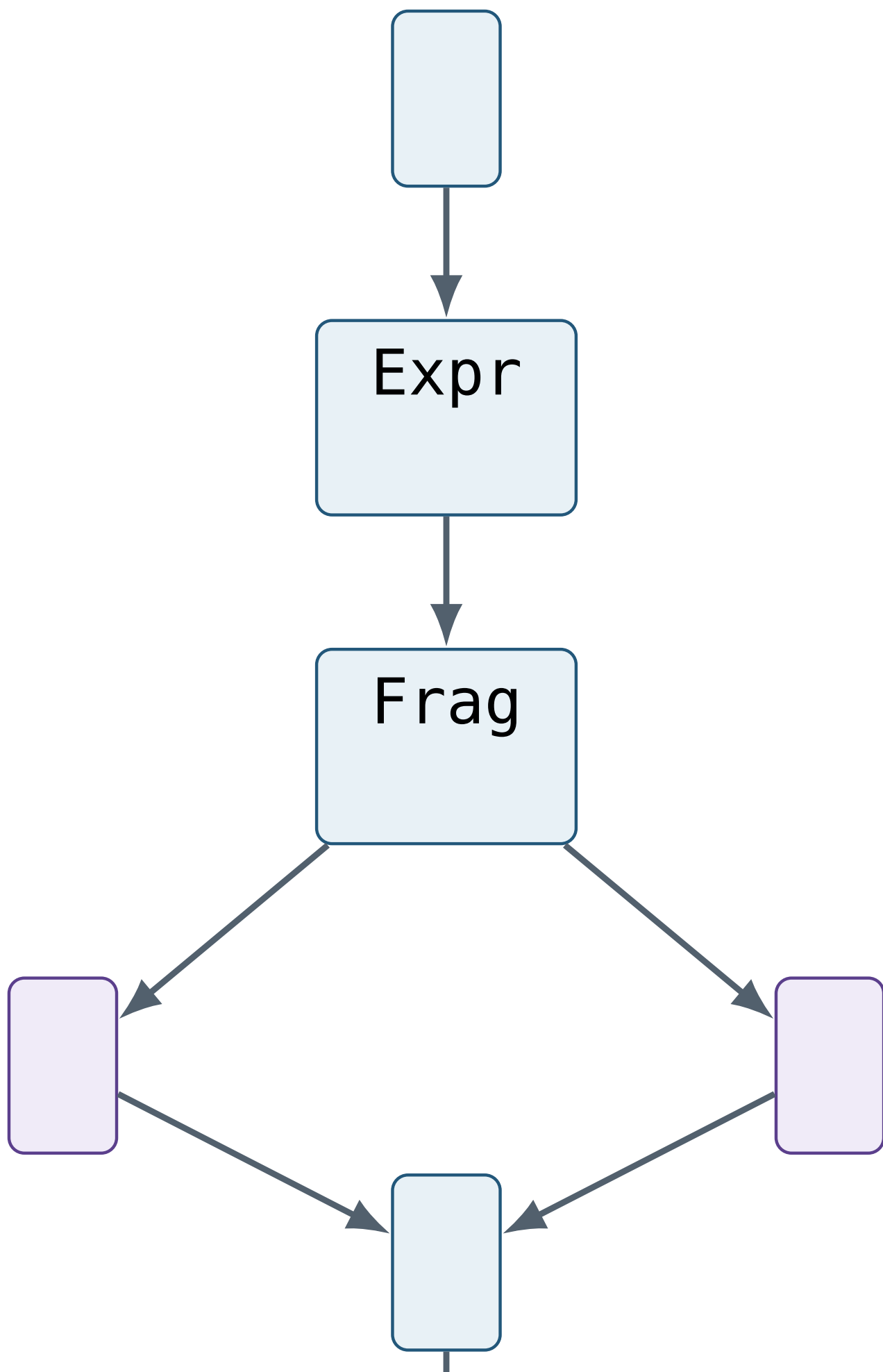
bv_decide

verifyCert_correct

→ → →

→ → .

MVarId



```
set_option autoImplicit true in
noncomputable example : (goal) := (candidate)
```

ExprFrag

```
data SynthOutcome
  = SynthCandidates [[String]] [String]
  | SynthRefuted Bool
  | SynthNoTerm [String]
```

SynthRefuted
satunsatunknown

ProdSum



sat

unknown

unknown

unsatsatunknown

```
(declare-const use_h0 Bool)
(declare-const use_h1 Bool)
(assert (=> use_h0 H0))
(assert (=> use_h1 H1))
(check-sat-assuming (use_h0 use_h1 goal_negated))
(get-unsat-core)
```

grind

unsafePerformIO

unknown

grind
omegaliagrind
bv_decide
grind

grindunsat

grobnerunknown

```
##print axioms
```

```
refine
```

$$\begin{array}{l} \Gamma \vdash TP \\ Pt \end{array}$$

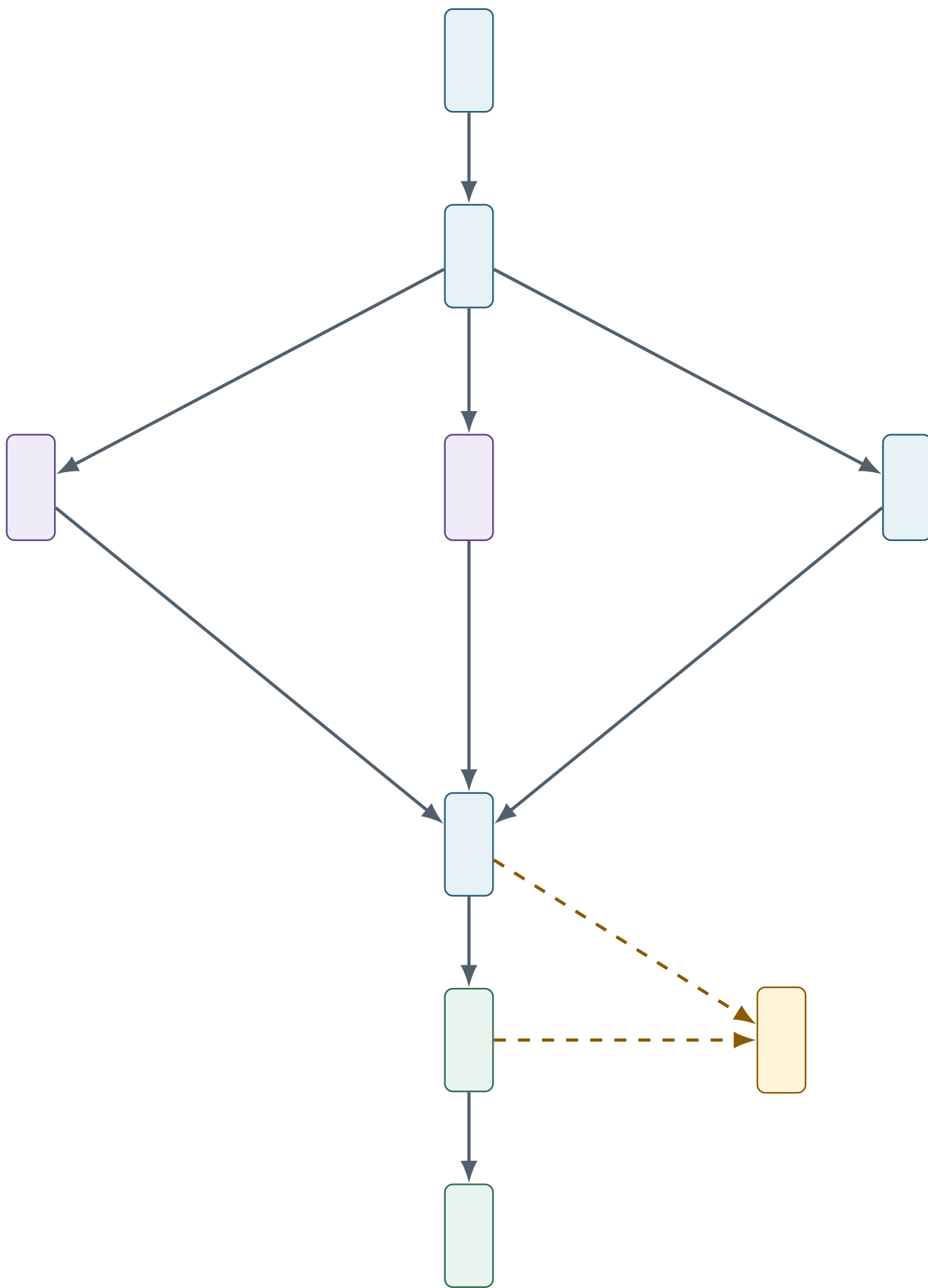
$$\Gamma(t) : T \implies \Gamma \vdash t : T.$$

$$(\Gamma \vdash T) \implies P(\Gamma \wedge \neg T).$$

```
structure ProjectionReport where
  exactConnectives      : Array SourceId
  opaqueAtoms           : Array SourceId
  erasedTypeclassBinders : Array SourceId
  monomorphizedDecls     : Array Name
  approximations         : Array Approximation
  depthTruncations       : Array SourceId
  unsupportedTerms       : Array UnsupportedTerm
  modelRealization       : ModelRealizationLevel
  negativeAdequacy       : NegativeAdequacyLevel
```

Boolcomplete

unknown



```
structure GoalSnapshot where
```

```
  snapshotId      : UInt64
  target          : Expr
  localContext    : LocalContext
  localInstances  : LocalInstances
  levelParams     : Array Name
  metavariableCtx : MetavarContext
  transparency    : TransparencyMode
  environmentHash : ByteArray
  sourceMap       : SourceMap
  options         : Options
  dependencySlice : Array Name
```

SourceIdExpr

```
structure ExternalCandidate where
```

```
  snapshotId : UInt64
  candidateId : UInt64
  payload     : CandidatePayload
  provenance  : Provenance
  projection  : ProjectionReport
  evidence    : ExternalEvidence
  ranking     : RankingData
  progress    : ProgressReport
```

CandidatePayload

```
inductive ProofNode
```

```
| exact      (term : TermRef)
| intro      (binder : BinderRef) (body : NodeId)
| apply      (function : TermRef) (arguments : Array NodeId)
| constructor (ctor : Name) (fields : Array NodeId)
| cases      (scrutinee : FVarRef) (branches : Array CaseBranch)
| have       (type : TypeRef) (proof body : NodeId)
| rewrite     (rule : TheoremRef) (direction : Bool) (body : NodeId)
| induction  (major : FVarRef) (motive : TypeRef)
              (branches : Array CaseBranch)
| tactic     (capability : Capability) (inputFacts : Array FactRef)
| hole       (holeId : HoleId) (expected : TypeRef)
              (context : ContextSlice) (capabilities : CapabilitySet)
```

```
structure ProofGraph where
  nodes      : Array ProofNode
  root       : NodeId
  sharing    : Array SharingConstraint
  sourceMap  : SourceMap
```

TermRef

omegagrind

```
inductive ReconstructionResult
| closed   (proof : Expr) (audit : AuditTrail)
| refined  (proof : Expr) (goals : Array MVarId) (audit : AuditTrail)
| rejected (reason : ReconstructionError) (audit : AuditTrail)
```

TryThisexactrefine

SourceId

grind

h1h4h7h1h4grindh7

grind

leant?

leant

leant (engine := djinn|exference|both)

leant_trace

solve?

MVarId

leant-server

sorry

TryThis

Expr

```
{
  "schema": 1,
  "snapshot": "9f38...",
  "leanRevision": "1d2dde6d64cb...",
  "leantRevision": "0160237e4bdf...",
  "djexRevision": "f50b5eac8672...",
  "goal": { "sort": "prop", "fragment": { ... } },
  "providers": [ ... ],
  "premises": [ ... ],
  "engine": "both",
  "limits": {
    "candidateWindow": 60,
    "verificationWindow": 24,
    "exferenceSteps": 128
  }
}
```

LibrarySuggestions"leant"

example

`debug.skipKernelTC`

Expr

have

$(\gamma, \gamma, \gamma, \gamma, \gamma)$.

grind

```
by
  intro f h x
  refine f x ?_
  -- goal: index x < bound
```

omegaomeagrind

exact

introrefinecases

solveWithZ3

findCounterexample

findUnsatCore

proposeInstances

solveTemplate

checkBranchFeasible

proposeInvariant

obtainProofTrace

```
structure SmtQuery where
  logic      : SmtLogic
  declarations : Array SmtDecl
  assertions  : Array TrackedAssertion
  assumptions : Array AssumptionLiteral
  sourceMap   : SmtSourceMap
  timeoutMs   : Nat
  rlimit      : Option Nat
  randomSeed  : UInt32
  requestModel : Bool
  requestCore  : Bool
  requestProof : Bool
  strategy     : SolverStrategy
```

Bool

BitVec

Fin n

AbstractModel

grind

```
structure InstantiationPlan where
  theorem      : TheoremRef
  typeArgs     : Array TermRef
  explicitArgs : Array TermRef
  triggerTerms : Array SmtTermId
  origin       : InstantiationOrigin
  dependencies : Array FactRef
```

grind

$$a_0 + a_1x_1 + \cdots + a_nx_n \geq 0$$

Feasible

Infeasible

Unknown

pushpop

grind

grind

grind

```
inductive EqReason
| hypothesis (fact : FactRef)
| symmetry   (reason : ReasonId)
| transitivity (left right : ReasonId)
| congruence  (function : TermRef) (arguments : Array ReasonId)
| rewrite     (theorem : TheoremRef) (arguments : Array TermRef)
```

omega

liagrind

linarithnlinarith

grind

bv_decide

bv_decide

bv_decidegrindgrindbv_decide

cases

selectstore

unknown

grobner

grind

```
structure TheoryAdapter where
  name      : Name
  recognizes : GoalSnapshot -> MetaM Recognition
  project    : GoalSnapshot -> Recognition -> MetaM SmtQuery
  validateModel : SmtModel -> MetaM ModelValidation
  reconstruct : UnsatEvidence -> MetaM ReconstructionResult
  capabilities : CapabilitySet
  version     : Nat
```

grindTryThis

TryThis

grind

bv_decide

bv_decidegrind

GrindMgrind

grind

grind

LibrarySuggestions

grind

solve?

```
example (n m : Nat) (h1 : n <= m) (h2 : m < n + 1) : n = m := by
  solve?
-- Candidate lanes:
--   omega: accepted
--   leant + omega skeleton: accepted, larger
--   z3 core -> omega: accepted, same premises h1 h2
-- Try this:
--   omega
```

grind

UnknownNoInstanceWithinBounds

grind

$$x \rightarrow x' a_i$$

$$\rho(x) = a_0 + \sum_i a_i x_i, \qquad \rho(x) \geq 0, \qquad \rho(x') < \rho(x).$$

grind

```
structure SnapshotId where value : UInt64
structure SourceId   where value : UInt32
structure BinderId   where value : UInt32
structure TermId     where value : UInt32
structure TypeId     where value : UInt32
structure FactId     where value : UInt32
structure NodeId     where value : UInt32
structure HoleId     where value : UInt32
```

Name

Expr

```
{
  "capabilities": {
    "proofGraph": 2,
    "residualHoles": true,
    "rankNBinders": 6,
    "exactProviderAssignments": true,
    "structuralProdSumKinds": true,
    "smt": ["QF_UF", "QF_LIA", "QF_BV", "QF_AUFBV"],
    "modelValidation": ["finite", "bitvector", "integer"],
    "evidence": ["candidate", "core", "model", "instancePlan"]
  }
}
```

```
inductive TruncationReason
| timeout
| heartbeatLimit
| memoryLimit
| candidateLimit
| choicePointLimit
| queuePruned (count : Nat)
| binderLimit (observed supported : Nat)
| providerLimit
| instanceLimit
| depthLimit
| solverUnknown (reason : String)
| interrupted
```

```
structure ProgressReport where
```

```
phase      : Name
explored   : Nat
candidates : Nat
elapsedMs  : Nat
truncation : Array TruncationReason
completeFor : Array Capability
```

completeFor

```
inductive ExternalEvidence
| candidateOnly
| structuralDerivation (derivation : DerivationGraph)
| structuralRefutation (refutation : RefutationGraph)
                      (adequacy : NegativeAdequacyLevel)
| model (model : SmtModel) (validation : ModelValidationLevel)
| unsatCore (facts : Array FactId) (minimality : CoreMinimality)
| instantiations (plans : Array InstantiationPlan)
| theoryPlan (plan : TheoryProofPlan)
| certificate (format : CertificateFormat) (bytes : ByteArray)
| proofTrace (format : ProofTraceFormat) (bytes : ByteArray)
| invariantTemplate (template : TemplateAssignment)
| noEvidence
```

certificateproofTrace


```

namespace ProofPortfolio

structure ReconstructConfig where
  allowResidualGoals : Bool := true
  maxResidualGoals   : Nat  := 8
  verifyKernel       : Bool := true
  axiomPolicy        : AxiomPolicy := .report

opaque snapshotGoal : MVarId -> MetaM GoalSnapshot
opaque projectForLeant : GoalSnapshot -> MetaM LeantRequest
opaque projectForSmt   : GoalSnapshot -> TheoryAdapter -> MetaM SmtQuery

opaque reconstructGraph :
  GoalSnapshot -> ProofGraph -> ReconstructConfig -> MetaM ReconstructionResult

opaque validateModel :
  GoalSnapshot -> SmtQuery -> SmtModel -> MetaM ModelValidation

opaque replayTheoryPlan :
  GoalSnapshot -> TheoryProofPlan -> MetaM ReconstructionResult

opaque addTryThisSuggestion :
  GoalSnapshot -> ReconstructionResult -> TacticM Unit

end ProofPortfolio

```

```

data SearchRequest = SearchRequest
{ requestSnapshot  :: SnapshotId
, requestGoal      :: TypedFragment
, requestProviders :: [Provider]
, requestPremises  :: [Premise]
, requestCapabilities :: CapabilitySet
, requestLimits    :: SearchLimits
}

data SearchEvent
= CandidateEvent CandidateGraph Provenance RankingData
| DiagnosticEvent Diagnostic
| ProgressEvent ProgressReport
| FinishedEvent Completion

runSearch :: SearchRequest -> ConduitT () SearchEvent IO ()

```

grind

pushpop

leant?
leant
z3?
counterexample?
solve?
explain_failure

$x = 0, y = 1$
 $U \neq \emptyset$

```
solve? (using := [h1, h2, theoremName])
      (without := [expensiveLemma])
      (engines := [grind, leant, z3])
      (maxPremises := 24)
      (allowInduction := false)
      (counterexamples := .finite 8)
```

xmap_preserveshhBoundhNonnegomega

```
by
  induction xs with
  | nil => simp
  | cons x xs ih =>
    refine constructor
    . exact ?length_bound
    . simp using ih
```

leant?

TryThis

Exprrefine

unknownunknown

grind

omegabv_decide

unsat

solve?

grind

LibrarySuggestions

`simpgrindomegagrobnerbv_decide`

`grind`

`bv_decide`

`Expr`

sorry

PropBool

grind

unsat

leant?

refine

grindunknown

by_decidegrindgrind

satunsatunknown

Leant.Tactic

MVarId

sorry

leant?TryThisleantleant_trace

grind

leant?

grind
grind

omegaliagrind
bv_decide bv_decide
grind
grind

grind

grobner

leant?

omega

TryThisrefine

omega

BitVec 8

example

LibrarySuggestions

grind

grind

leanprover/lean4@1d2dde6d64cb
src/Lean/Elab/Tactic/Grind.lean
src/Lean/Elab/Tactic/Grind/Main.lean
src/Lean/Meta/Tactic/Grind/Extension.lean
src/Lean/Meta/Tactic/Grind/Types.lean
src/Lean/Meta/Tactic/TryThis.lean
src/Lean/LibrarySuggestions/Basic.lean
src/Std/Tactic/BVDecide/Reflect.lean
commit87e7204456bf:bv_decide/grindintegration
commitf29e9e488ea8:state-dependentterminationmeasures

VladimirReshetnikov/Leant@dab7a1518f51
latestfunctionalLeantintegration@0160237e4bdf
Leant/README.md
Leant/src/Leant/Synth/Fragment.hs
Leant/src/Leant/Synth/Engine.hs
Leant/src/Leant/Synth/Replay.hs
Leant/src/Main.hs
commit0160237e4bdf:structuralhigher-kindedsynthesisissupport
commitde4f562c77ff:six-binderDjexintegration
VladimirReshetnikov/Djex@f50b5eac8672
Djex/synthesis/.../Query.hs
Djex/synthesis/.../Search.hs
Djex/synthesis/.../Candidate.hs
Djex/synthesis/.../Generated.hs
Djex/synthesis/.../Selection.hs
Djex/djinn/.../Djex/Djinn.hs
Djex/exference/.../Djex/Exference.hs

Z3Prover/z3@0bd679a404be

z3/README.md

z3/src/api/z3_api.h

z3/src/api/z3_optimization.h

z3/src/api/z3_fixedpoint.h

z3/examples/python/proofreplay.py

commitdf8d23960eda:MBPlinearityfix

commit7d67dbfcfc81:quantified-realQERegression

commit5bd485eb03fe:polymorphismdependent-typeguard

commit0105b220fd18:cross-platformhashdeterminism

commit09aaadf96380:ASTconstructiondeterminism

ufmg-smite/lean-smt@a4ed29b0b9fd

lean-smt/README.md